

Air University



COAL (Lab)

Year 2025

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Academic Year: 2024 – 2028

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BS – Cyber Security – Fall – 24 (A)

Date of Submission: December 16, 2025

Lab Task # 13

Write an assembly language program to count the number of 1-bit (set bits) in the 16-bit register BX using the ROL instruction. Store the result in AX and display the count as a single digit on the screen.

Code

```
.MODEL SMALL
.STACK 100H
.DATA
    ; No data needed
.CODE
MAIN PROC
    MOV     BX, 0A7CCH    ; Load test value into BX

    MOV     AX, 0         ; Init count in AX
    MOV     CX, 16        ; Set loop for 16 bits

COUNT_LOOP:
    ROL     BX, 1         ; Rotate BX left
    JNC     NO_CARRY      ; Skip if no carry
    INC     AX            ; Inc count if carry
NO_CARRY:
    LOOP    COUNT_LOOP    ; Loop 16 times

    ; AX has bit count
    ; Display count (0-9)
    ADD     AL, '0'       ; To ASCII
    MOV     DL, AL        ; To DL
    MOV     AH, 2         ; DOS char output
    INT     21H           ; Display

    ; Exit
    MOV     AH, 4CH
    INT     21H

MAIN ENDP
END MAIN
```

Explanation:

- Initialization:** The program starts by setting up registers. It loads 'BX' with the value '0A7CCH' and initializes 'AX' to '0' (for counting bits) and 'CX' to '16' (the number of bits in 'BX').
- Bit Rotation (ROL):** The instruction 'ROL BX, 1' rotates the bits in 'BX' to the left by one position. The bit that gets pushed out of the leftmost position goes into the Carry Flag (CF) and becomes the new rightmost bit.
- Counting Logic:** The 'JNC NO_CARRY' instruction checks the Carry Flag. If there is no carry (the CF is '0'), it means a '0' bit was pushed out, so the count doesn't increase. If there is a carry (the CF is '1'), it means a '1' bit was pushed out, and the counter in 'AX' is incremented.

4. **Loop Control:** The `LOOP COUNT\_LOOP` instruction decrements `CX` and repeats the counting until `CX` reaches zero, ensuring all `16` bits in `BX` are checked.
5. **Result Display:** Once the loop is done, the final count is in the `AL` register. The instruction `ADD AL, '0'` converts the count to its ASCII form (like turning `9` into `9`). Finally, it displays the character using the DOS function `02H`.

Output

For input: 0A7CCH

```
D:\>D:\test
9
D:\>[
```

For input: 0AB0CH

```
D:\>D:\test
7
D:\>[
```

THE END